Discrete-time Controllers

Control Engineering

Course Schedule

1. Systems
2. Continuous-time Controllers
3. Laplace Domain
4. Poles/Zeros & Stability
5. Controller Design
6. Discrete-time Controllers
7. State Space Representation
8. Advanced Control Methods

Discrete-time Controllers

- Digital controllers
- Sampling & Zero-Order Hold
- Discrete-time modeling
- z-transfer functions
- Discretization methods

Why Digital Control?

Most modern controllers are implemented on:

- Microcontrollers
- DSPs (Digital Signal Processor)
- Computers

Advantages:

- Flexibility (software changes)
- Cost efficiency
- Advanced algorithms possible

Disadvantages:

- Sampling effects
- Quantization
- Delays (latency)

Continuous vs Digital Control

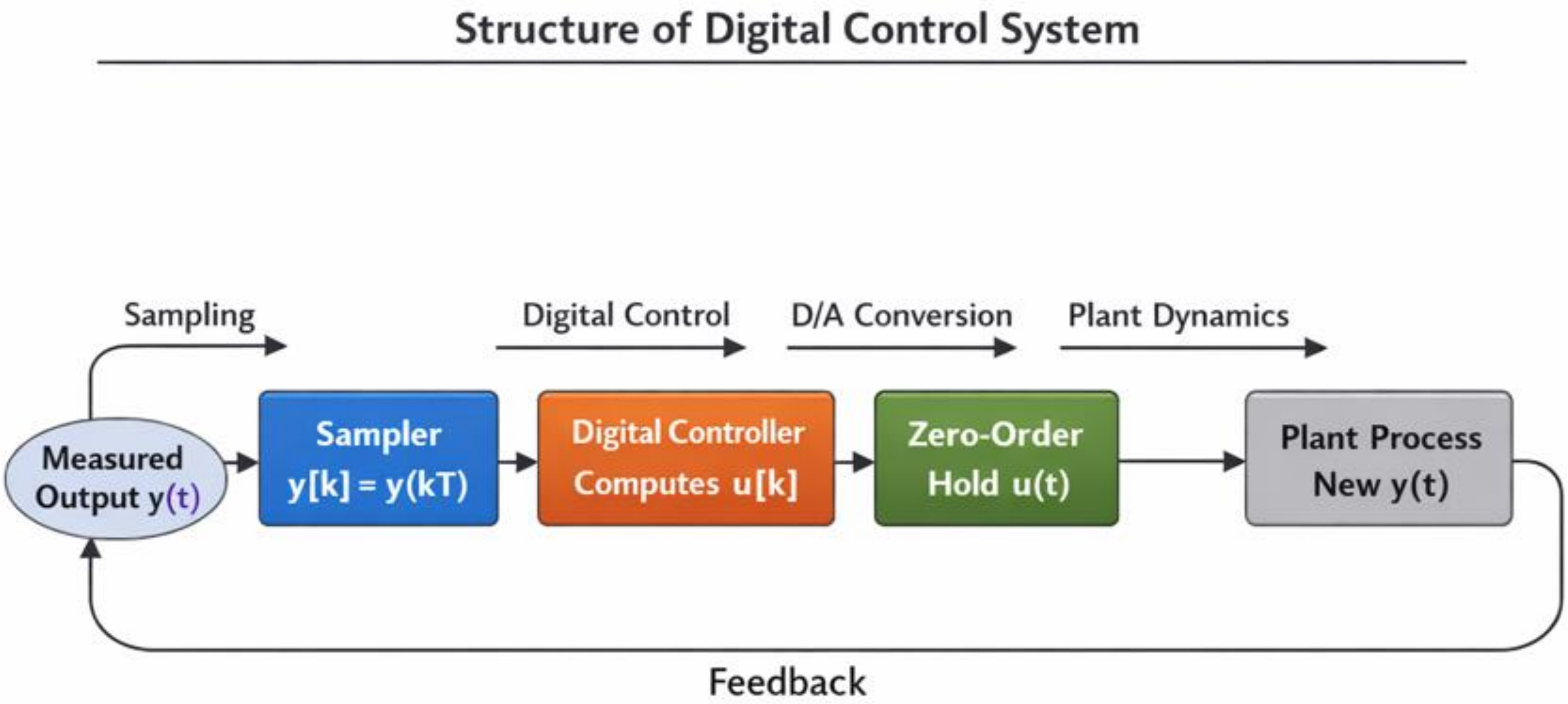
Continuous-time:

- Signals defined $\forall\, t \in \mathbb{R}$
- Differential equations

Digital (discrete-time):

- Signals defined at $t = kT$
- Difference equations

Key idea: approximation of continuous systems using sampled data



Idea:

controlling a continuous physical system using a digital computer

Sampling

Signals measured at discrete times only

$$x[k] = x(kT)$$

- T: sampling period
- k: integer index

Sampling frequency:

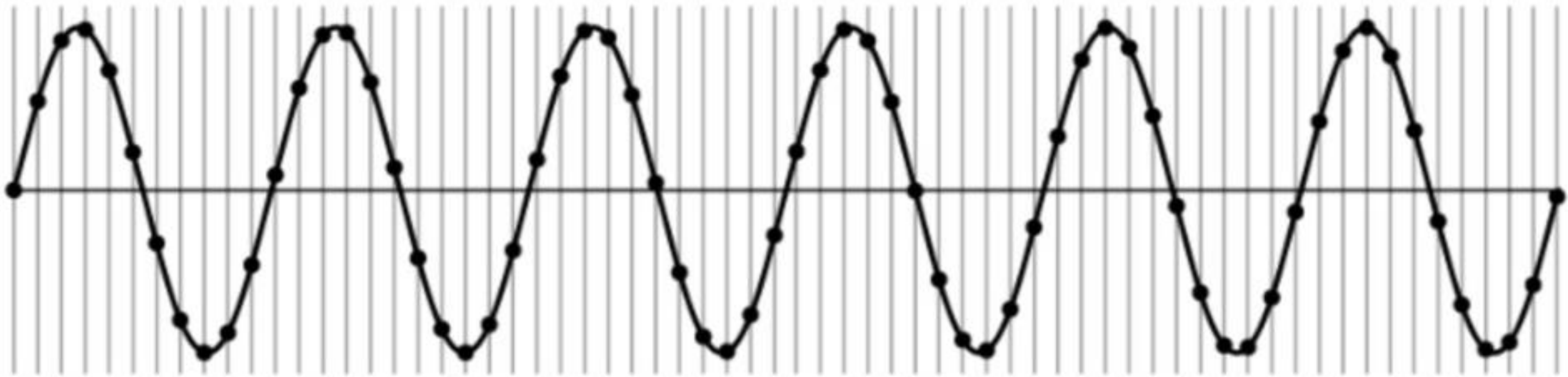
$$f_s = \frac{1}{T}$$

Angular frequency:

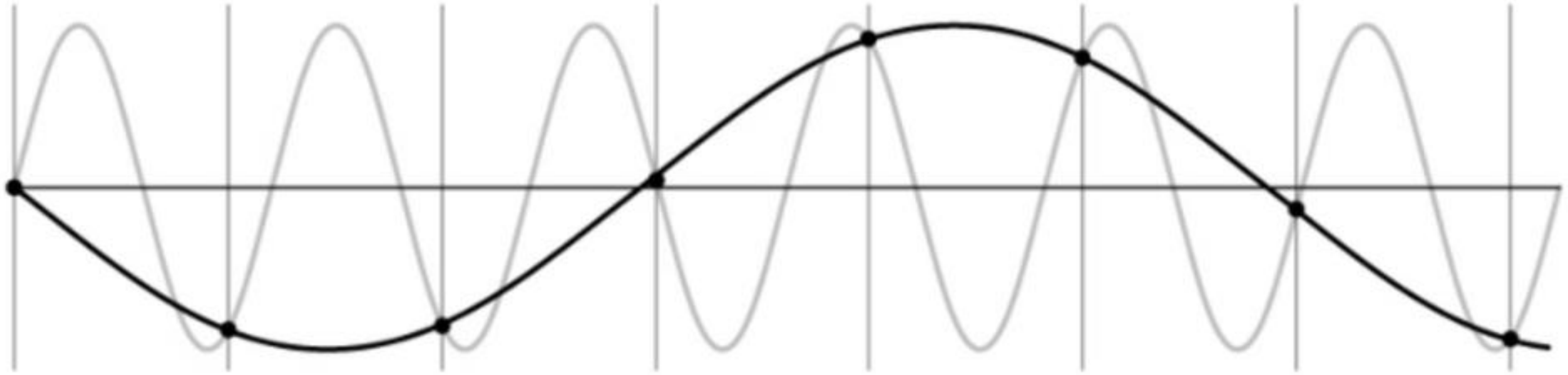
$$\omega_s = \frac{2\pi}{T}$$

Example:
$x(t) = \sin t$

→ $x[k] = \sin kT$



Undersampling



Aliasing: higher-frequency signal disguised as one of lower frequency

Example: Moiré Effect in Digital Images

Aliasing in digital images often manifests itself as ringing patterns

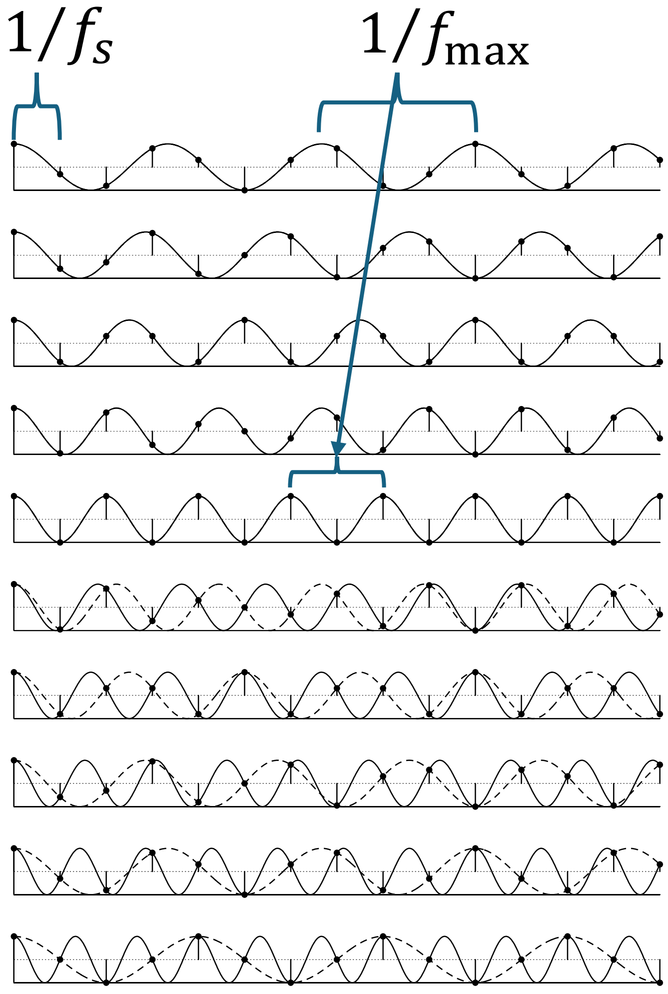
Temporal aliasing can be observed as wagon-wheel effect (apparent backward motion) in image sequences (video): frame rate too low with respect to speed of wheel rotation

Nyquist-Shannon Sampling Theorem

No aliasing (continuous signal can be recovered from its samples) if:

$$f_s > \underbrace{2f_{\max}}_{\text{Nyquist frequency}}$$

→ Must sample at least twice the highest signal frequency



Aliasing for:

$$\frac{1}{f_{\max}} < 2 \cdot \frac{1}{f_s}$$

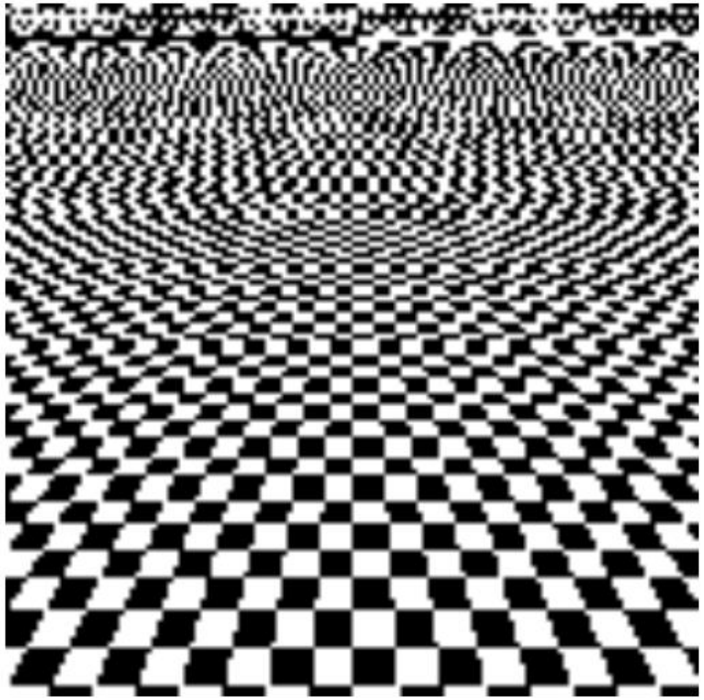
Digital images: no aliasing if $\frac{N}{2} > f_{\max}$, with number of pixels N

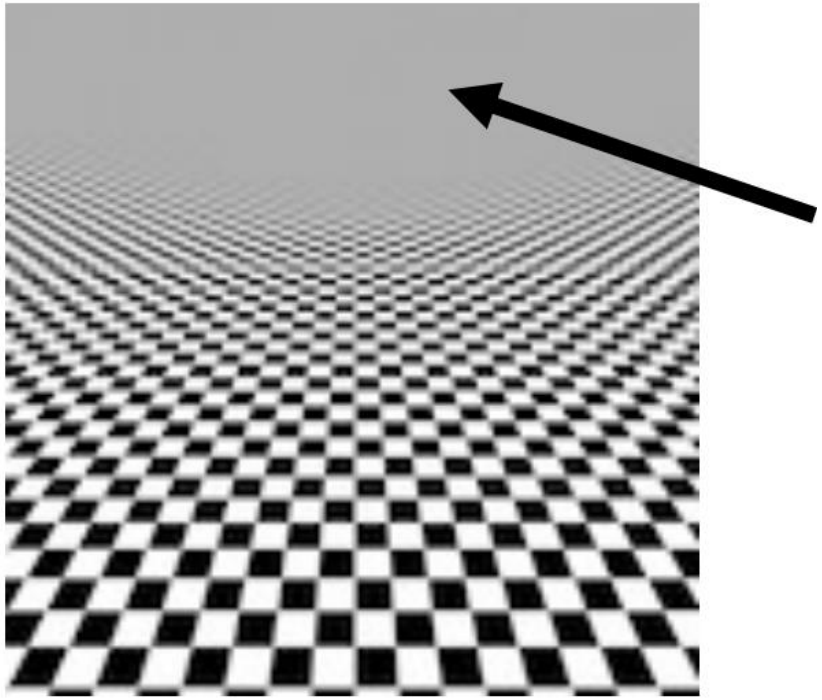
(second dimension accordingly)

Anti-Aliasing Filter

Solution: use a low-pass filter (smoothing) before sampling

→ remove signal frequencies above Nyquist frequency

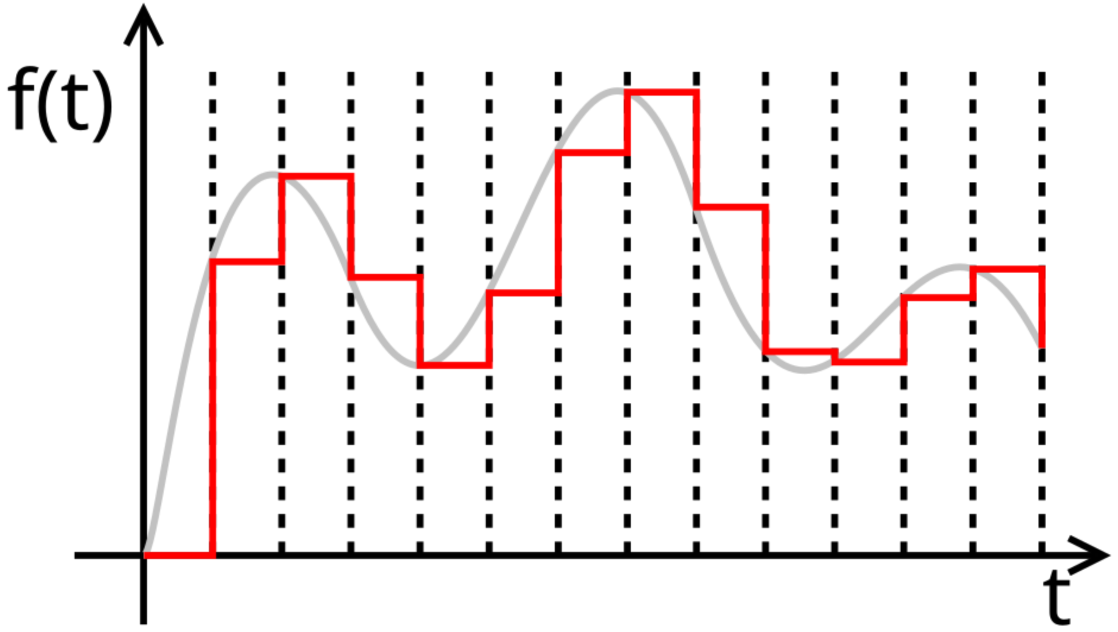




average of black and white

Zero-Order Hold (ZOH)

Holds input constant between samples
$u(t) = u[k], \qquad t \in [kT, (k+1)T)$



Real actuators receive continuous signals

→ digital output must be converted

ZOH models Digital-to-Analog Converter (DAC) behavior

Signals are now sequences

ZOH introduces:

- Delay
- Approximation error

From Differential to Difference Equations

Continuous-time systems: differential equations

Discrete-time systems ($t \to kT$): difference equations

Sampling a continuous-time system leads to a difference equation:

- Start with differential equation
- Solve over one sampling interval
- Evaluate at next sample

Example: First-Order System

$$\dot{y}(t) = -y(t) + u(t) \quad \text{with initial condition } y(0)$$

Constant input (ZOH):

$$\dot{y}(t) = -y(t) + u[k]$$

Homogenous solution: $\dot{y}(t) = -y(t)$

$$y_h(t) = C\, e^{-t}$$

Particular solution: try constant solution $y_p(t) = A$ (constant input $u[k]$)

$$0 = -A + u[k] \qquad \rightarrow \qquad y_p(t) = u[k]$$

General solution:

$$y(t) = y_h(t) + y_p(t) = C\, e^{-t} + u[k]$$

Initial condition:

$$y(0) = C + u[k] \quad \rightarrow \quad C = y(0) - u[k]$$

$$\rightarrow \quad y(t) = \underbrace{(y(0) - u[k])e^{-t}}_{\text{transient}} + \underbrace{u[k]}_{\text{steady state}}$$

- System moves from $y(0)$ to $u[k]$
- Exponential transition

$$\rightarrow \quad y(t) = e^{-t}y(0) + (1 - e^{-t})u[k]$$

Sample at $t = T$:

$$y((k+1)T) = e^{-T}y(kT) + (1 - e^{-T})u[k]$$

Discrete-time system: difference equation

$$y[k+1] = e^{-T}y[k] + (1 - e^{-T})u[k]$$

Difference Equations

First-order system template:

$$y[k+1] = \underbrace{a\, y[k]}_{\text{memory}} + \underbrace{b\, u[k]}_{\text{input effect}}$$

$a = e^{-\lambda T}$ comes from continuous-time physics

Second-order system template:

$$y[k+2] + a_1\, y[k+1] + a_2\, y[k] = b_0\, u[k] + b_1 u[k-1]$$

Time-shifted version to make it causal (current output from past values):

$$k \to k-2$$

$$y[k] + a_1\, y[k-1] + a_2\, y[k-2] = b_0\, u[k] + b_1 u[k-1]$$

Motivation for z-Transform

- How to solve difference equations?
- How to analyze stability?
- How to design controllers?

Continuous-time systems:

differential equation → Laplace transform → algebra (transfer function)

Discrete-time systems:

difference equation → z-transform → algebra (z-transfer function)

z-Transform Definition

$$X(z) = \sum_{k=0}^{\infty} x[k] z^{-k}$$

Laplace transform:

$$X(s) = \int_{0}^{\infty} x(t) e^{-st} \, dt$$

- $x[k]$: discrete-time signal
- z: complex variable

Interpretation:

- Converts a sequence into a function of z
- Discrete-time equivalent of the Laplace transform

Time Shift Property

$$\mathcal{Z}\{x[k+1]\} = zX(z)$$
$$\mathcal{Z}\{x[k-1]\} = z^{-1}X(z)$$
$$\mathcal{Z}\{x[k+n]\} = z^{n}X(z)$$

→ Shifts in difference equations ($y[k+1],\ y[k],\ y[k-1]$) become simple multiplications in the z-domain

z-Transfer Functions

Example:

$$y[k+1] = a\, y[k] + b\, u[k]$$

Apply z-transform:

$$z\, Y(z) = a\, Y(z) + b\, U(z)$$

Transfer Function:

$$G(z) = \frac{Y(z)}{U(z)} = \frac{b}{z-a}$$

Discrete equivalent of Laplace transfer functions

s-Plane Mapped to z-Plane

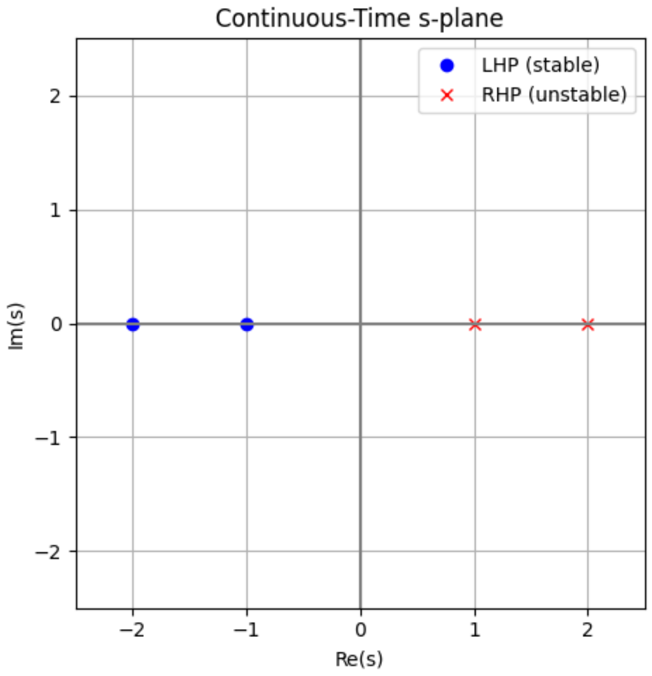
Sampling exponentials:

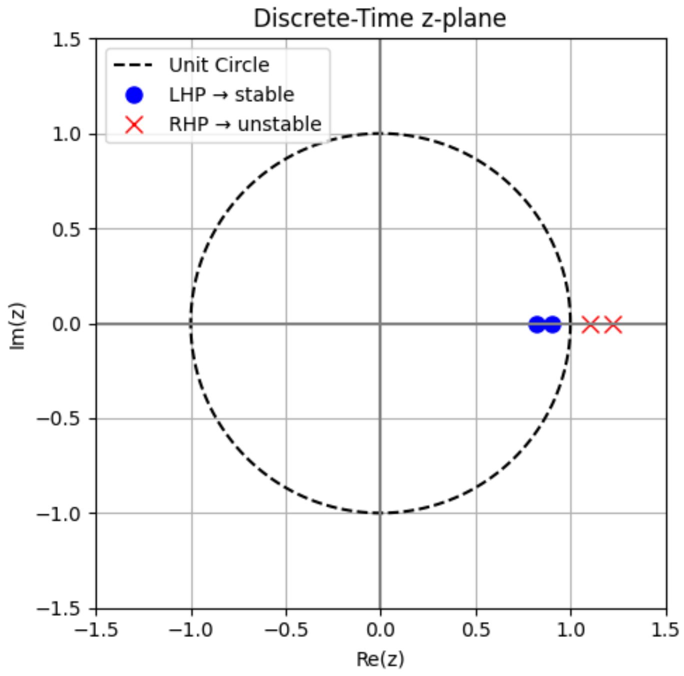
$$x(t) = e^{st}$$

$$x[k] = e^{skT} = (e^{sT})^k = z^k$$

→ Relation: $z = e^{sT}$

$s = \sigma + j\omega$

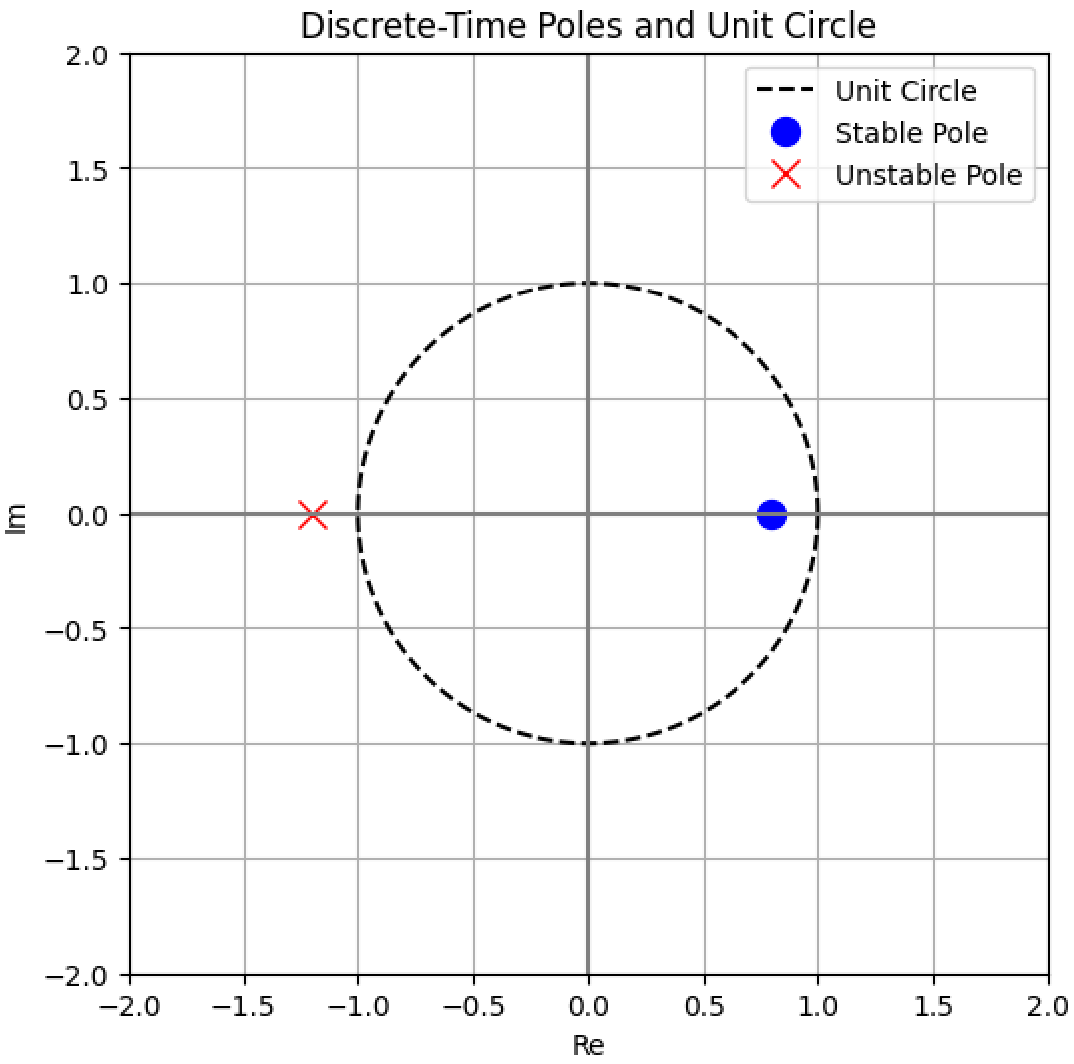




$z = e^{\sigma T} + e^{j\omega T}$

$\sigma = 0$ → unit circle

Stability in Discrete-Time Systems



Remember:

- $\mathrm{Re}(s_p) < 0$: stable
- $\mathrm{Re}(s_p) = 0$: marginally stable
- $\mathrm{Re}(s_p) > 0$: unstable

→ Discrete-time system stable if poles z_p lie inside unit circle

ZOH Equivalent Discretization

Goal: direct mapping $u[k] \to y[k]$

Steps: $u[k] \xrightarrow{\text{ZOH}} u(t) \xrightarrow{G(s)} y(t) \xrightarrow{\text{sampling}} y[k]$

ZOH modifies input as seen by the plant

ZOH equivalent discretization:

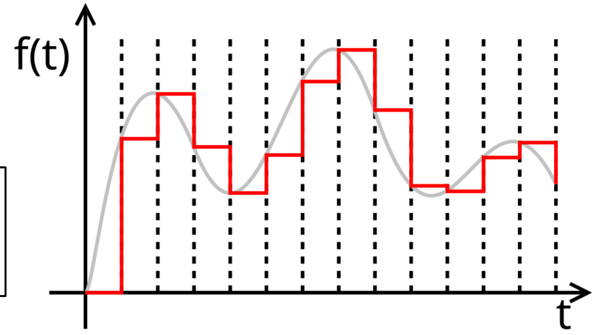
Discrete-time system that produces exactly the same sampled output as the continuous-time plant under ZOH input

ZOH Equivalent Transfer Function

Converts a continuous-time plant $G(s)$ into a discrete-time equivalent:

$$G(z) = \mathcal{Z}\left\{G(s) \cdot \underbrace{\frac{1-e^{-sT}}{s}}_{G_{\mathrm{ZOH}}(s)}\right\}$$

$$\boxed{1 - u(t-T) = \begin{cases} 1, & 0 < t < T \\ 0, & t \geq T \end{cases}}$$



input is sequence of **rectangular pulses**: step responses over one interval

- $\frac{1}{s}$ corresponds to constant signal — turn ON at $t = 0$
- e^{-sT} corresponds to delay by T → delayed constant: $\frac{e^{-sT}}{s}$ — turn OFF (−) at $t = T$

ZOH Discretization Example

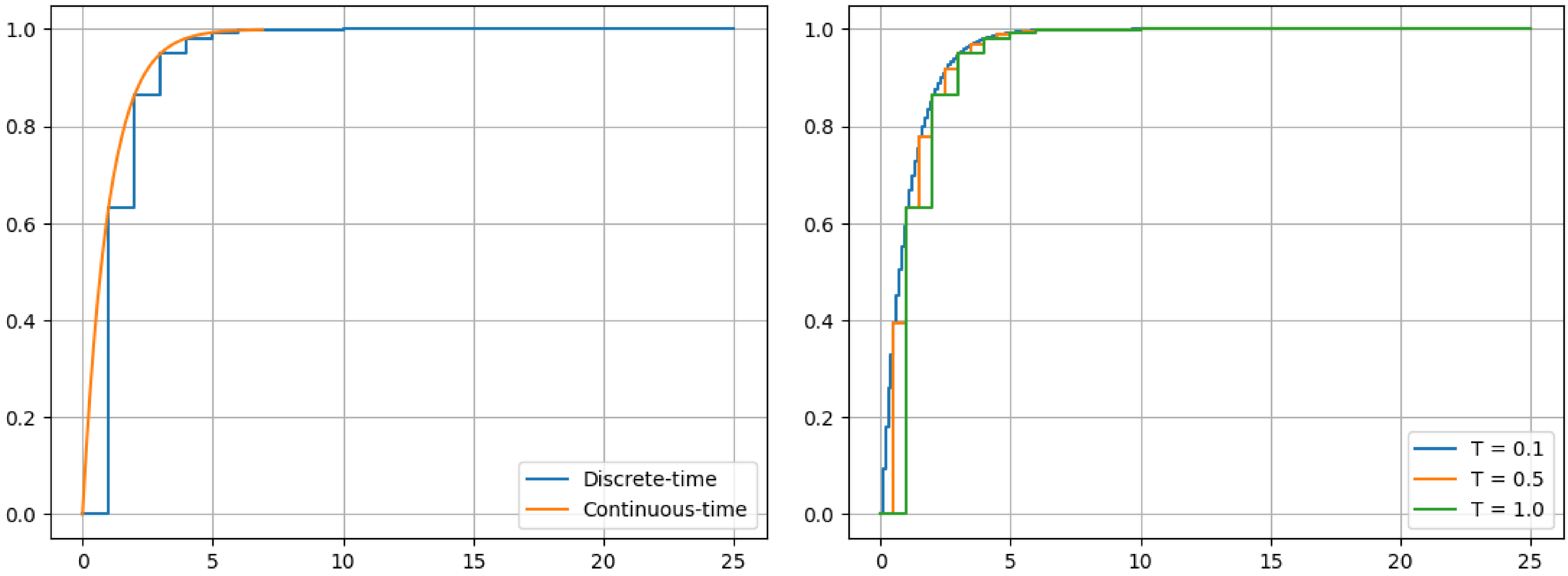
Given continuous-time system:

$$G(s) = \frac{1}{s+1}$$

Goal: Find discrete-time transfer function, using ZOH with sampling time T

1. Apply ZOH: $G(s) \cdot \frac{1-e^{-sT}}{s} = \frac{1}{s+1} \cdot \frac{1-e^{-sT}}{s}$
2. Go to time domain: $y(t) = \mathcal{L}^{-1}\left\{\frac{1}{s+1} \cdot \frac{1-e^{-sT}}{s}\right\} = (1 - e^{-t}) - \left(1 - e^{-(t-T)}\right)u(t-T)$
3. Sample output: $y[k] = y(kT) = e^{-(k-1)T}(1 - e^{-T}), \quad k \geq 1$
4. Take z-transform: $G(z) = \mathcal{Z}\{y[k]\} = \frac{1-e^{-T}}{z-e^{-T}}$

Step Response Comparison



Alternative Discretization Methods

How can I approximate a continuous transfer function in discrete time without modeling ZOH explicitly?

Goal:

$G(s) \rightarrow G(z)$ (approximative)

Methods:

- Forward Euler
- Backward Euler
- Tustin (bilinear)

Forward Euler Method

$$s \approx \frac{z-1}{T}$$

Derivation:

Approximation of slope using forward difference:

$$\dot{y}(t) = \frac{dy}{dt} \approx \frac{y(t+T)-y(t)}{T} = \frac{y[k+1]-y[k]}{T}$$

Consider first-order system (for example): $\dot{y}(t) = -y(t) + u(t)$

→ $\frac{y[k+1]-y[k]}{T} = -y[k] + u[k]$ → $y[k+1] = (1-T)y[k] + Tu[k]$ (explicit calculation)

→ $zY(z) = (1-T)Y(z) + TU(z)$ → $G(z) = \frac{Y(z)}{U(z)} = \frac{T}{z-(1-T)}$

Compare $G(z) = \frac{T}{z-(1-T)}$ with $G(s) = \frac{1}{s+1}$ → $s \approx \frac{z-1}{T}$

Forward Euler Stability

Compare with ZOH for first-order system:

ZOH: $G(z) = \frac{1-e^{-T}}{z-e^{-T}}$ Euler: $G(z) = \frac{T}{z-(1-T)}$

→ Euler ≈ ZOH for small sampling time: $e^{-T} \approx 1 - \mathrm{T}$

Consider stable continuous system with $s_p = -1$

→ Euler mapping: $z = 1 - T$

Stability requires: $|1 - T| < 1$ → $0 < T < 2$

→ Euler can make a stable system (the model, not the physical process!) unstable if T is too large

Backward Euler

$$s \approx \frac{z-1}{Tz}$$

Backward difference instead of future slope approximation (forward Euler):

$$\dot{y}(t) \approx \frac{y[k+1]-y[k]}{T} \rightarrow \dot{y}(t) \approx \frac{y[k]-y[k-1]}{T}$$

First-order system: $\frac{y[k]-y[k-1]}{T} = -y[k] + u[k] \rightarrow (1+T)Y(z) = z^{-1}Y(z) + TU(z)$

$\rightarrow G(z) = \frac{T}{(1+T)-z^{-1}} = \frac{Tz}{(1+T)z-1}$

Implicit solution instead of explicit calculation of $y[k+1]$

Pole location compared to ZOH: $e^{-T} \approx \frac{1}{1+T}$

$\frac{1}{1+T} < 1$ for all $T > 0$ (always inside unit circle) → stable, but numerical damping (implicit)

Tustin (Bilinear) Method

$$s \approx \frac{2}{T} \cdot \frac{z-1}{z+1}$$

Maps entire LHP inside unit circle

Series development and termination after the first term

Derivation:

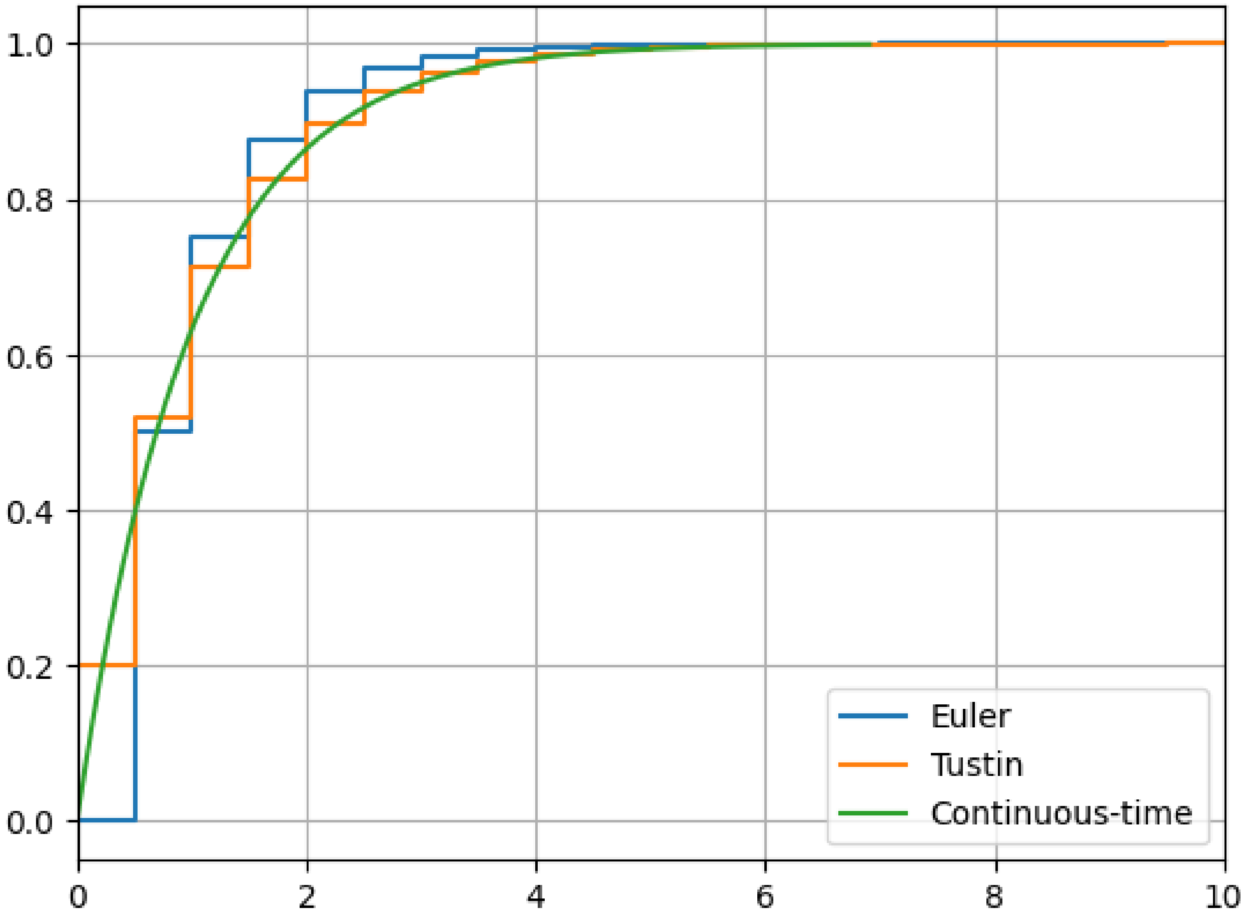
$$z = e^{sT} \;\rightarrow\; s = \frac{1}{T} \ln z \approx \frac{1}{T} \cdot 2 \cdot \left[\frac{z-1}{z+1} + \cdots\right]$$

First-order system: $G(z) = \frac{1}{\frac{2}{T} \cdot \frac{z-1}{z+1} + 1} = \frac{T}{2} \cdot \frac{2(z+1)}{2(z-1) + T(z+1)} = \frac{\frac{T}{2}(z+1)}{z - 1 + \frac{T}{2} z + \frac{T}{2}} = \frac{\frac{T}{2}(z+1)}{\left(1 + \frac{T}{2}\right) z - \left(1 - \frac{T}{2}\right)}$

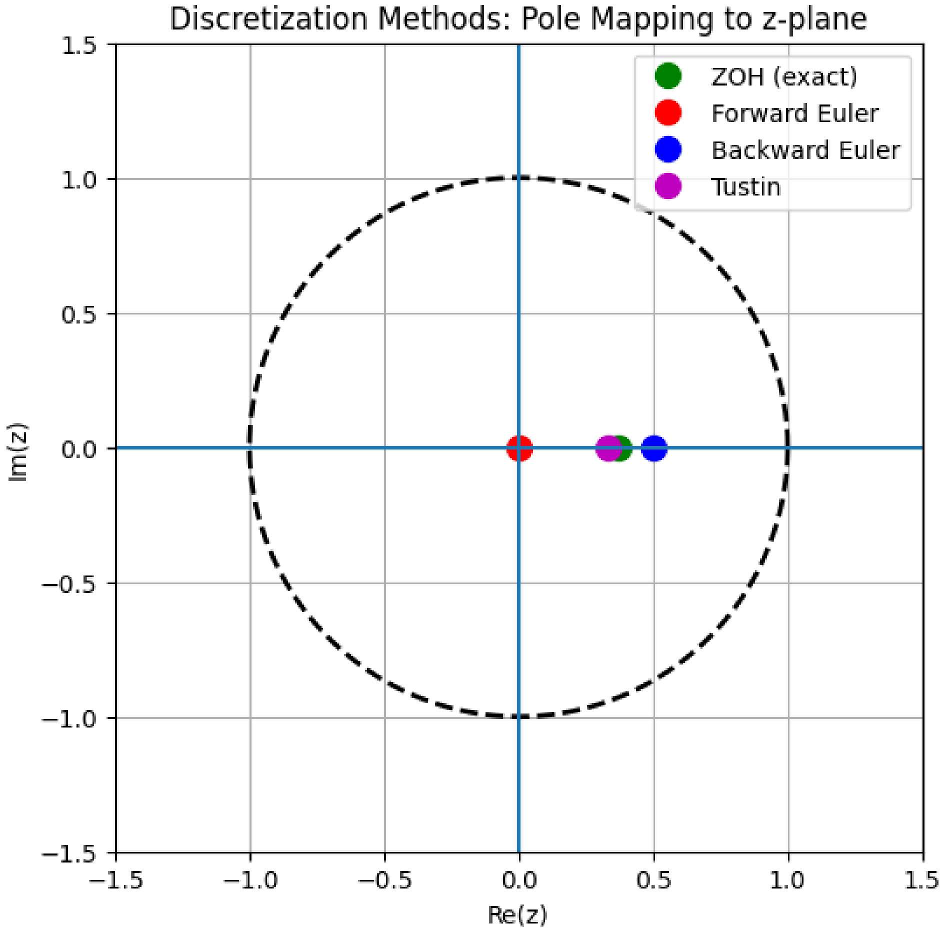
Corresponds to bilinear transformation, for example $G(s) = \frac{1}{s}$:

$$y[k] = y[k-1] + T \cdot \frac{u[k] + u[k-1]}{2}$$

Comparison



1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
Euler
Tustin
Continuous-time



Discretization Methods: Pole Mapping to z-plane
ZOH (exact)
Forward Euler
Backward Euler
Tustin
Im(z)
Re(z)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5

Implementation Issues

- Quantization: finite precision, rounding errors
- Computation delay: controller takes time to compute → must compute within sampling period
- Sampling time trade-offs:
    - Small T → accuracy ↑, computation ↑
    - Large T → accuracy ↓

Discrete-Time PID Tuning

PID controller (continuous-time):

$$u(t) = K_p\, e(t) + K_i \int e(t)\, dt + K_d\, \frac{de(t)}{dt}$$

In discrete-time:

$$u[k] = K_p\, e[k] + K_i\, T \sum e[k] + K_d\, \frac{e[k]-e[k-1]}{T}$$

Position form

Key differences from continuous PID:

- Signals are sampled: $e[k] = e(kT)$
- Integration becomes summation
- Derivative becomes difference
- Numerical approximation required

Incremental Form

Position form uses full sum → accumulates error, sensitive to numerical issues

Idea: use $\Delta u[k] = u[k] - u[k-1]$

$$u[k-1] = K_p\, e[k-1] + K_i\, T \sum e[k-1] + K_d\, \frac{e[k-1] - e[k-2]}{T}$$

→ $\Delta u[k] = K_p\, (e[k] - e[k-1]) + K_i\, T\, e[k] + K_d\, \frac{e[k]-2e[k-1]+e[k-2]}{T}$

→ $u[k] = u[k-1] + \Delta u[k]$

No explicit sum
Control updates instead of absolute values

PID Transfer Function

Continuous PID transfer function:

$$G_c(s) = K_p + \frac{K_i}{s} + K_d s$$

$$A = K_p + \frac{K_i\, T}{2} + \frac{2\, K_d}{T}$$

$$B = K_i\, T - \frac{4\, K_d}{T}$$

$$C = -K_p + \frac{K_i\, T}{2} + \frac{2\, K_d}{T}$$

Discrete PID via Tustin $\left(s \approx \frac{2}{T} \cdot \frac{z-1}{z+1}\right)$:

$$G_c(z) = K_p + \frac{K_i\, T\, (z+1)}{2\, (z-1)} + \frac{K_d\, 2\, (z-1)}{T\, (z+1)} = \cdots = \frac{A\, z^2 + B\, z + C}{z^2 - 1} = \frac{U(z)}{E(z)}$$

→ $z^2\, U(z) - U(z) = A\, z^2\, E(z) + B\, z\, E(z) + C\, E(z)$

→ $u[k+2] - u[k] = A\, e[k+2] + B\, e[k+1] + C\, e[k]$ $\qquad k \to k-2$

→ $u[k] = u[k-2] + A\, e[k] + B\, e[k-1] + C\, e[k-2]$

Direct Discrete Tuning

Option 1: Tune directly in discrete-time

Option 2: Tune continuous PID → discretize

Most common: second approach

For example:

- Ziegler–Nichols
- Convert to discrete using Tustin

Filtered Derivative

Problem: $\frac{e[k]-e[k-1]}{T}$ amplifies noise

→ Use: $D(s) = \frac{K_d\, s}{1+\tau\, s}$

Then apply Tustin for discrete-time

Summary

- Digital control is essential in modern systems
- Sampling introduces constraints
- ZOH connects discrete and continuous worlds
- z-transform is key analysis tool
- Discretization method important for controller design